

KINTO Smart Ops

Warehouse, Delivery Challan & Vendor-Warehouse System

Comprehensive Implementation Plan

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Executive Summary

This document provides a comprehensive implementation plan for a multi-warehouse inventory system with delivery challans, vendor-warehouse linking, consignment management, and integration with existing invoice/dispatch workflows in the KINTO Smart Ops platform.

1. Introduction

1.1 What is a Delivery Challan?

A Delivery Challan (DC) is a GST-compliant document for goods movement without immediate sale. Unlike invoices, it does not create tax liability and uses triplicate copies. It is required for stock transfers, job-work, samples, and returnable goods.

1.2 Challan vs Invoice

- Invoice: Creates tax liability, used for sales
- Delivery Challan: No tax liability, used for goods movement
- DC can be linked to invoice or standalone

1.3 Warehouse Concept

Warehouses are locations where inventory is stored. They can be company-owned (factory, depot) or vendor-hosted (distributor, consignment agent). The system tracks stock levels, movements, and enables inter-warehouse transfers.

2. Delivery Challan - Invoice Linkage

2.1 Invoice-First Flow

Standard sales flow where invoice is created first, then DC is generated from invoice lines for dispatch tracking.

- Invoice Created !' DC Generated !' Goods Dispatched !' POD Received

2.2 DC-First Flow

Field sales flow where goods are dispatched first with DC, then converted to invoice later.

- DC Created !' Goods Dispatched !' Customer Confirms !' Invoice Generated

2.3 Standalone DC (No Invoice)

Used for internal movements, samples, job-work where no sale occurs.

- Stock Transfers between warehouses
- Samples and exhibitions
- Job-work material dispatch
- Returnable goods

3. Delivery Challan Types

Type	Purpose	Invoice Link
Sale	Delivery against invoice	Required
Returnable	Samples, demo units	Optional
Job Work	Goods sent for processing	No
Stock Transfer	Inter-warehouse movement	No
Sample	Free samples	No
Exhibition	Trade show goods	No

4. Vendor-Warehouse Concept

4.1 Linked-Vendor Warehouse Model

Instead of converting vendors to warehouses, we link vendors to warehouse entities. This preserves vendor analytics while enabling stock tracking at vendor locations.

4.2 Use Cases

- Distributor holding consignment stock
- Stockist with company inventory
- Consignment agent selling on behalf
- Job-work partner processing materials
- External rented storage

4.3 Inventory Types at Vendor-Warehouse

- Consignment: Unsold stock (company ownership)
- Owned: Sold to vendor via invoice (vendor ownership)
- Reserved: Committed for pending orders

5. Database Schema Design

5.1 New Tables

- warehouses - Core warehouse master
- warehouse_vendor_relations - Vendor-warehouse linkage with terms
- warehouse_inventory - Stock by warehouse/product/batch
- stock_ledger - Movement history and audit trail
- delivery_challans - DC header with workflow status
- delivery_challan_items - DC line items
- stock_transfers - Inter-warehouse transfer tracking
- stock_transfer_items - Transfer line items
- consignment_settlements - Vendor settlement tracking
- invoice_delivery_links - DC-Invoice relationship

6. Workflow & State Machine

6.1 Delivery Challan States

- Draft - Initial creation
- Approved - Stock reserved, ready for dispatch
- In Transit - Goods dispatched, gatepass issued
- Delivered - POD received
- Returned - Goods returned (for returnable DC)
- Cancelled - DC cancelled

6.2 Stock Transfer States

- Pending - Request created
- Approved - Ready for dispatch
- In Transit - Goods moving
- Received - Verified and inventory updated
- Cancelled - Transfer cancelled

7. Feature Breakdown

Module 1: Warehouse Master

- Warehouse CRUD with ownership types
- Vendor linking capability
- GST location codes for compliance
- Default warehouse per user/role

Module 2: Warehouse Inventory

- Stock view by warehouse
- Stock ledger with movement history
- Min/Max stock alerts
- Batch and lot tracking
- Expiry date tracking

Module 3: Delivery Challans

- Full DC CRUD operations
- Generate DC from invoice
- Multiple challan types
- Workflow status management
- E-Way Bill integration
- Print template
- POD upload and tracking

Module 4: Stock Transfers

- Transfer request creation
- Approval workflow
- Paired DC generation (out/in)
- Receipt with variance handling

Module 5: Consignment Management

- Track consignment stock at vendors
- Vendor sales reporting
- Settlement calculation
- Settlement invoice generation

Module 6: Integration & Reports

- Update dispatch tracking dashboard
- Update gatepass flow
- Warehouse stock report
- DC register for GST
- Consignment aging report

8. Role Permissions

Screen Key	Label		Actions
warehouses	Inventory	Warehouse Master	create, edit, delete, view,
warehouse_inventory		Warehouse	view, create, edit
delivery_challans	Stock Transfers	Delivery Challans	view, create, edit, delete
stock_transfers		Stock Transfers	create, edit, view,
consignment_management	Consignment Settlements	Consignment Management	view, create, edit
consignment_settlements		Consignment Settlements	view, create

9. API Endpoints

Warehouses

- GET /api/warehouses - List all warehouses
- POST /api/warehouses - Create warehouse
- GET /api/warehouses/:id/inventory - Warehouse stock

Delivery Challans

- GET /api/delivery-challans - List challans
- POST /api/delivery-challans - Create challan
- POST /api/delivery-challans/:id/approve - Approve DC
- POST /api/delivery-challans/from-invoice/:id - Generate from invoice

Stock Transfers

- GET /api/stock-transfers - List transfers
- POST /api/stock-transfers - Create transfer
- POST /api/stock-transfers/:id/receive - Receive transfer

Consignment

- GET /api/consignment/summary/:warehouseId - Consignment overview
- POST /api/consignment/settlements - Create settlement

10. UI Screens

Screen	Location	Purpose
Warehouse List	Admin !' Warehouses	Manage all warehouses
Warehouse Detail	Warehouses/:id	View stock, ledger, vendors
Vendor Warehouses	Vendor !'	Linked locations
Delivery Challan List	Warehouses tab	
	Sales !' Challans	
Create DC	Sales !' New DC	All DCs with filters
DC from Invoice	Invoice !' Create DC	Create new challan
Stock Transfers	Inventory !' Transfers	Auto-generate DC
Consignment Dashboard	Sales !'	Inter-warehouse moves
	Consignment	Vendor stock overview
Settlements	Sales !' Settlements	Vendor settlements

11. Implementation Timeline

Week	Phase	Deliverables
Week 1	Foundation	Warehouse master, inventory tables, basic CRUD
Week 2	Inventory	Stock ledger, warehouse views, transfers
Week 3	Delivery Challans	DC CRUD, types, workflow, integration
Week 4	Consignment	Vendor-warehouse linking, tracking
Week 5	Integration	Dispatch tracking, gatepass, invoice linking
Week 6	Reports & Polish	Reports, print templates, testing

12. GST Compliance

12.1 Document Requirements

- Delivery Challan must have unique serial number
- Triplicate copies: Original, Duplicate, Triplicate
- GSTIN of consignor and consignee
- HSN codes for all items
- E-Way Bill for goods > Rs. 50,000

12.2 Tax Treatment by Movement Type

- Company !' Vendor Warehouse: DC (No tax - consignment)
- Vendor Warehouse !' Customer: Invoice (GST charged)
- Unsold Return: Return Challan (No tax)
- Settlement (Sale to Vendor): Invoice (GST charged)

End of Document

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